

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – CASE STUDY

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO3: Articulation (BL4, BL5)	Assessment:	Assessment Tool 2 – Case Study
Weightage:	20% of CIA	Total Marks:	100
CO3 Statement:	Design Divide and Conquer algorithms for Merge Sort, Quick Sort, Binary Search and Matrix Multiplication		
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Section / Batch:		Date:	

Instructions to Students

- Answer the following question. Total Marks: 100.
- Carefully analyze the given case study questions.
- Explain in detail with sample code if needed.
- Clearly identify all key issues, provide an in-depth analysis, and propose innovative, feasible solutions with strong justification.

Question Type	Focus Area	Questions
Case Study	Focus on Design Divide and Conquer algorithms: Merge Sort, Quick Sort, Binary Search and Matrix Multiplication	Q1

SECTION A – CASE STUDY

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

1. Case Study: Search Engine Query Processing (Binary Search)

Answer:

- A search engine stores millions of keywords in sorted order. The main goal is to retrieve the required keyword as quickly as possible. Since the data is already sorted, Binary Search is the best algorithm because it reduces the search space by half in every step.
- Binary Search follows the divide-and-conquer approach. It first checks the middle element. If the keyword is not found, it continues searching only in the left or right half based on the comparison. This process continues until the keyword is found.

Key Points:

1. Requires data to be sorted.
2. Uses the divide-and-conquer technique.
3. Time Complexity: $O(\log n)$.
4. Performs fewer comparisons than Linear Search.
5. Suitable for large datasets.

Solution & Justification:

Maintain the search engine index in sorted order and apply Binary Search whenever a user searches for a keyword. This provides fast retrieval, improves performance, and handles millions of records efficiently.

2. Case Study: Library Management Search System (Binary Search)

Answer:

- A digital library stores thousands of books in a sorted catalog. The objective is to help users find a required book quickly. Since the records are already sorted, Binary Search is an efficient solution for fast retrieval.
- Binary Search follows the divide-and-conquer technique by checking the middle book in the catalog. If the required book is not found, the search continues only in the left or right half until the correct record is located.

Key Points:

1. Catalog must be sorted.
2. Uses divide-and-conquer.
3. Time Complexity: $O(\log n)$.
4. Reduces the number of comparisons.
5. Suitable for large library databases.

Solution & Justification:

Store books in sorted order based on title or Book ID and apply Binary Search for searching. This provides quick access, improves search efficiency, and saves time for both students and library staff.

3. Case Study: Digital Dictionary Lookup System (Binary Search)

Answer:

- A digital dictionary stores words in alphabetical order. Users expect to find word meanings instantly, making Binary Search the most suitable algorithm for this task.
- The algorithm compares the target word with the middle word in the dictionary. If it is not found, the search continues only in the appropriate half of the remaining words. This process repeats until the required word is found.

Key Points:

1. Words must be stored alphabetically.
2. Uses divide-and-conquer.
3. Time Complexity: $O(\log n)$.
4. Faster than Linear Search.
5. Efficient for large dictionaries.

Solution & Justification:

Maintain the dictionary in sorted order and use Binary Search for every lookup. This provides fast word retrieval, reduces search time, and improves the overall performance of the application.

4. Case Study: Student Record Retrieval System (Binary Search)**Answer:**

- A university database stores student records in sorted order using registration numbers. The aim is to retrieve student information quickly and accurately whenever required.
- Binary Search applies the divide-and-conquer method by comparing the required registration number with the middle record. It repeatedly eliminates half of the remaining records until the correct student information is found.

Key Points:

1. Records should be sorted.
2. Uses divide-and-conquer.
3. Time Complexity: $O(\log n)$.
4. Provides fast and accurate retrieval.
5. Suitable for large student databases.

Solution & Justification:

Arrange student records in sorted order based on registration number and use Binary Search for searching. This ensures faster retrieval, improves database performance, and efficiently handles a large number of student records.

5. Case Study: Contact Search in Mobile Application (Binary Search)

Answer:

- A mobile application stores contacts in sorted order by name. Users expect to find contacts instantly, so Binary Search is an effective algorithm for quick searching.
- Binary Search follows the divide-and-conquer approach by checking the middle contact first. If the contact is not found, the search continues only in the relevant half of the contact list until the required contact is located.

Key Points:

1. Contacts must be sorted.
2. Uses divide-and-conquer.
3. Time Complexity: $O(\log n)$.
4. Requires fewer comparisons.
5. Suitable for mobile contact management.

Solution & Justification:

Maintain contacts in sorted order and use Binary Search whenever a user searches for a contact. This improves search speed, reduces response time, and provides an efficient user experience even with thousands of contacts.

RUBRICS FOR CALCULATION

Case Study					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25%	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25%	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20%	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20%	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10%	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25%				
Application of Concepts	25%				
Analysis	20%				
Solution Design	20%				
Clarity	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____